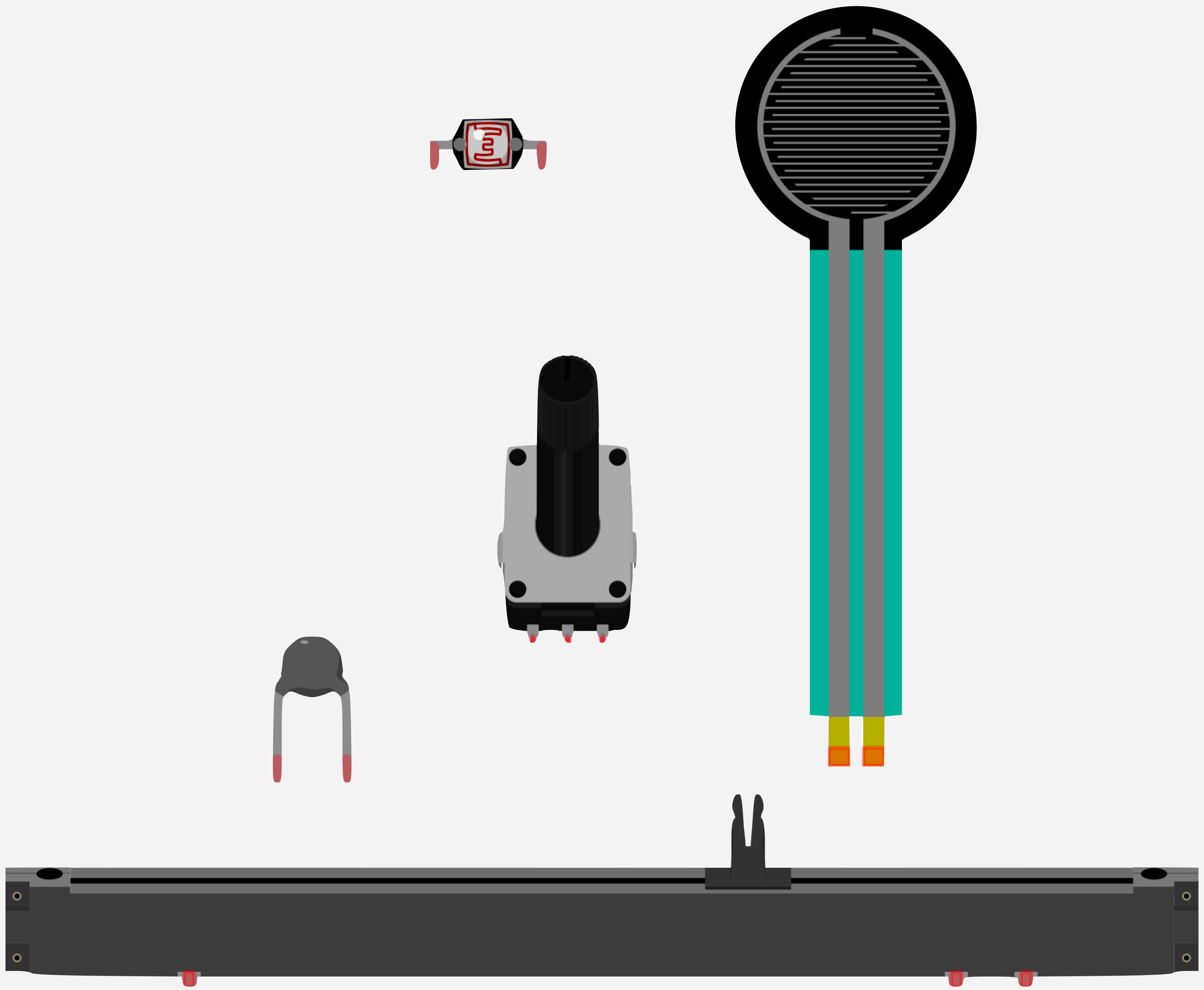
ANALOG INPUT

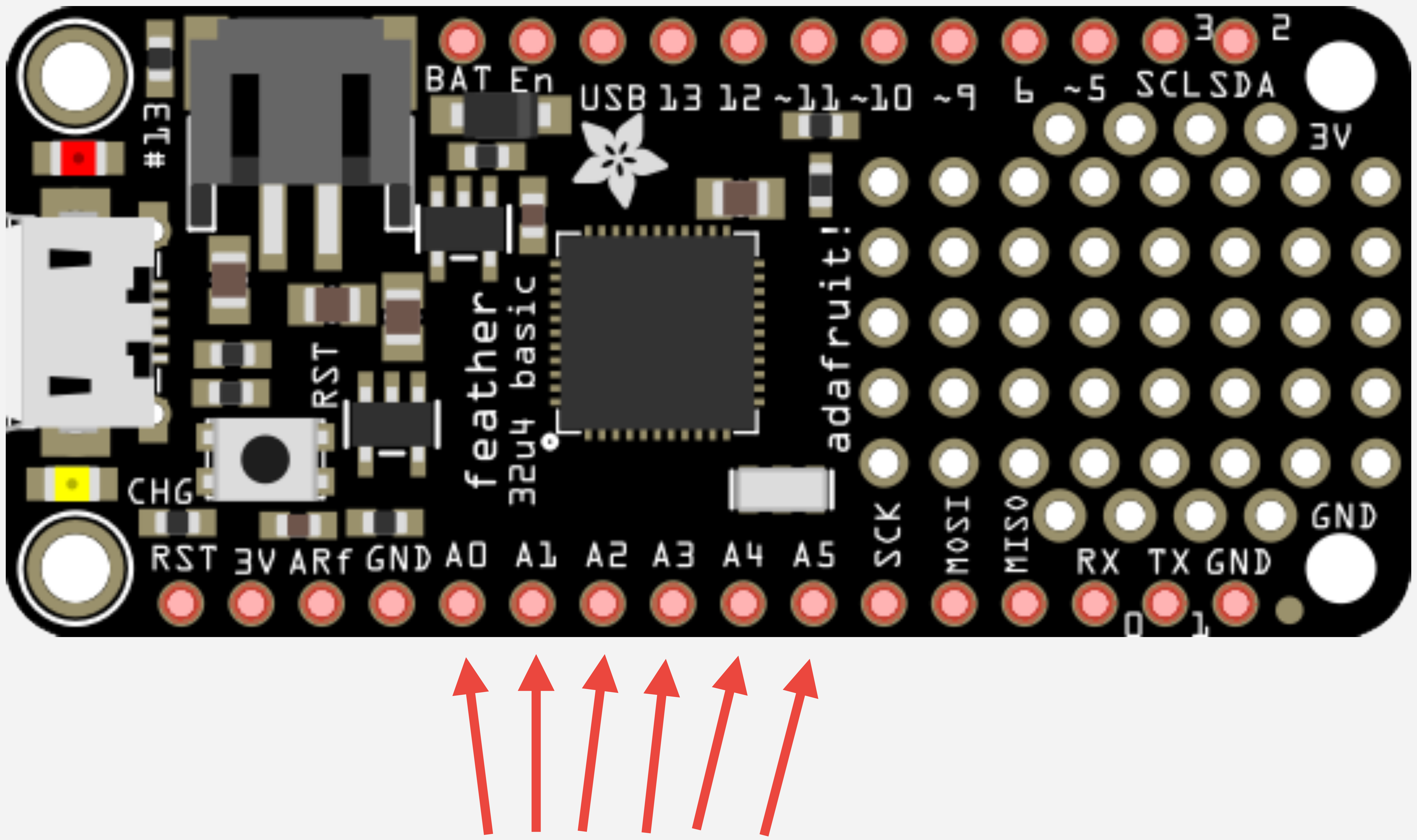


Knobs, Sliders, Temperature, Light, & more!

Analog Input

An analog input is any device that can create or transmit a range of values beyond just on/off.

Using Analog Inputs with the Arudino



Analog Input Pins

Analog Input

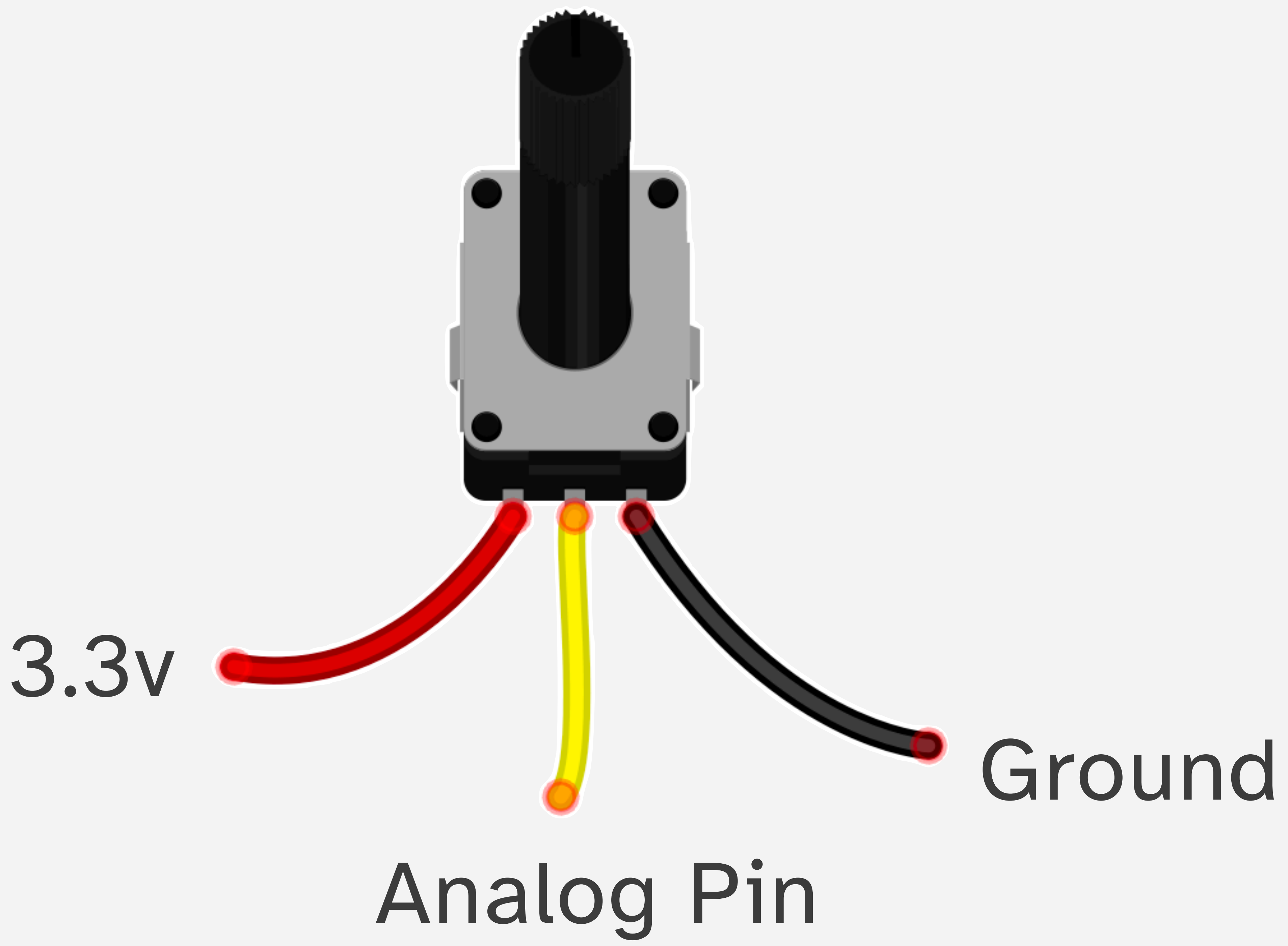
The inputs on the Arduino read voltage.

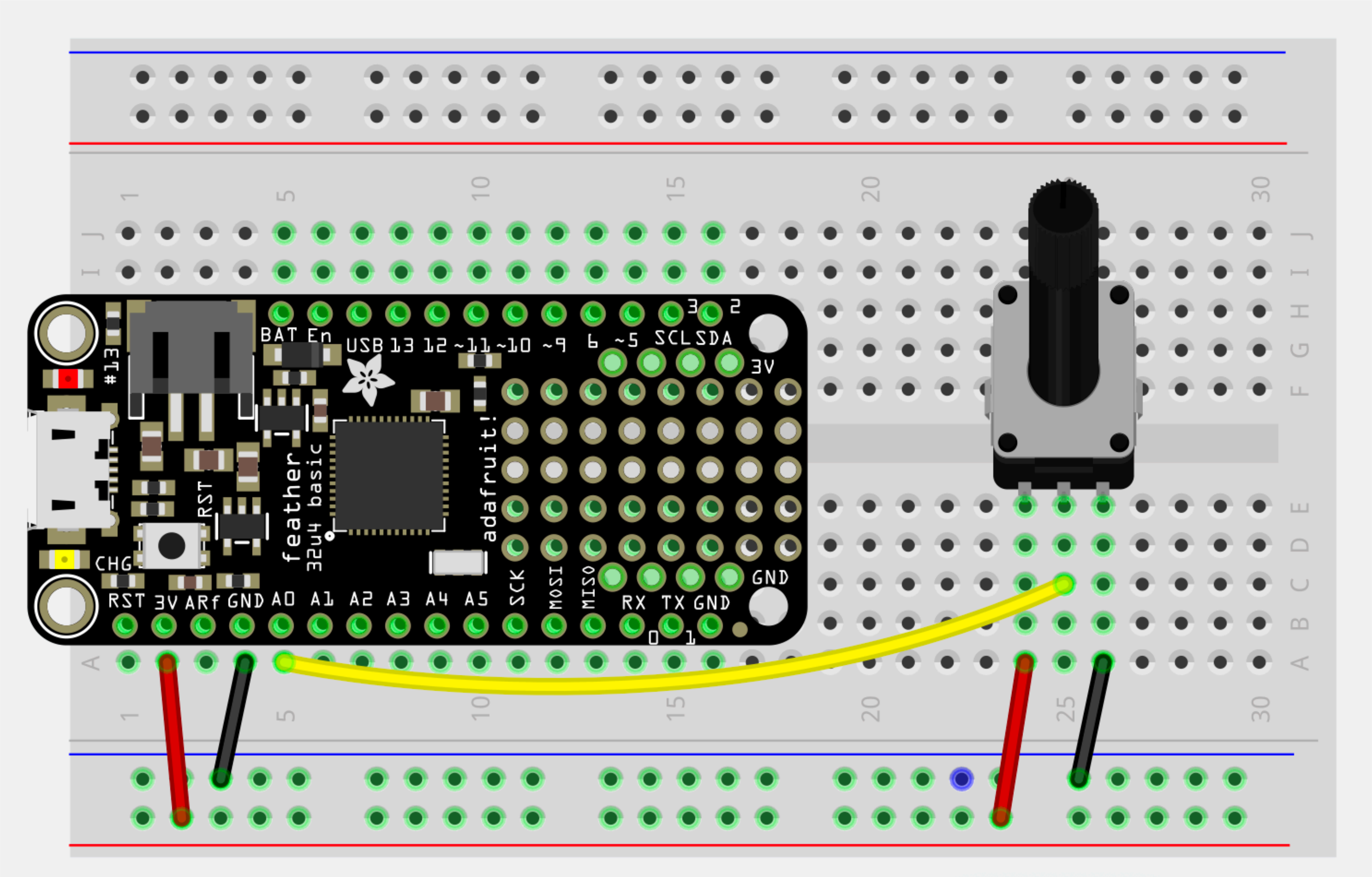
All inputs need to be thought of in terms of voltage differentials.

Unlike digital inputs, analog inputs convert differences in voltage levels into a number from 0-1024

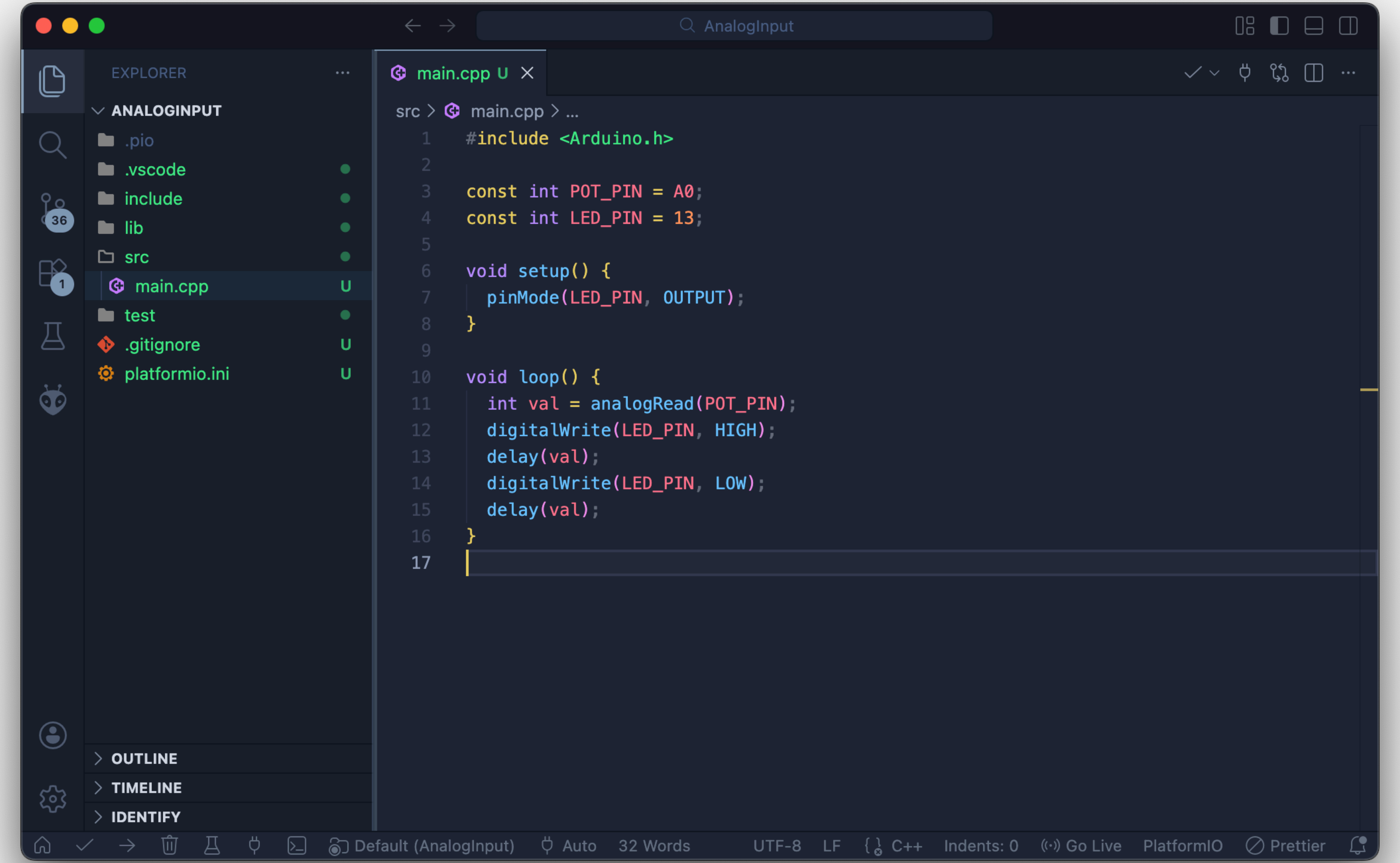
Analog Input

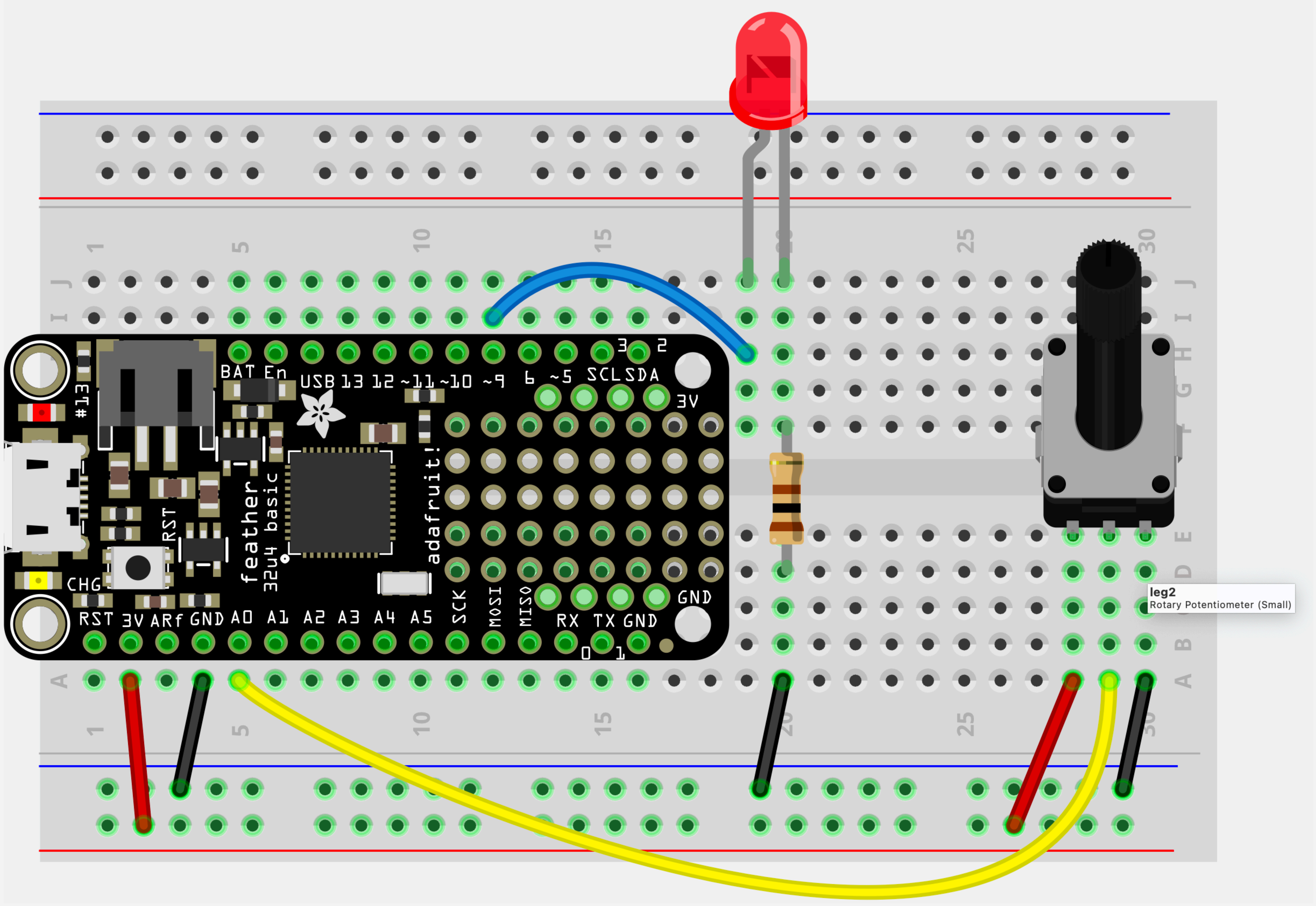
The potentiometer is the easiest analog input to use. It has three connections.





BAT En USB 13 12 ~11 ~10 ~9 6 ~5 SCL SDA 3 2 3V
#13
feather
32u4 basic
adafruit!
RST
CHG
RST 3V ARf GND A0 A1 A2 A3 A4 A5 SCK MOSI MISO RX TX GND
0 1
GND


AnalogInput
EXPLORER
ANALOGINPUT
.pio
.vscode
include
lib
src
main.cpp U
test
.gitignore U
platformio.ini U
OUTLINE
TIMELINE
IDENTIFY
main.cpp U
src > main.cpp > ...
#include <Arduino.h>
const int POT_PIN = A0;
const int LED_PIN = 13;
void setup() {
pinMode(LED_PIN, OUTPUT);
}
void loop() {
int val = analogRead(POT_PIN);
digitalWrite(LED_PIN, HIGH);
delay(val);
digitalWrite(LED_PIN, LOW);
delay(val);
}
Default (AnalogInput)
Auto
32 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier



BAT En USB 13 12 ~11 ~10 ~9 6 ~5 SCL SDA 3V
3 2
#13
CHG
RST
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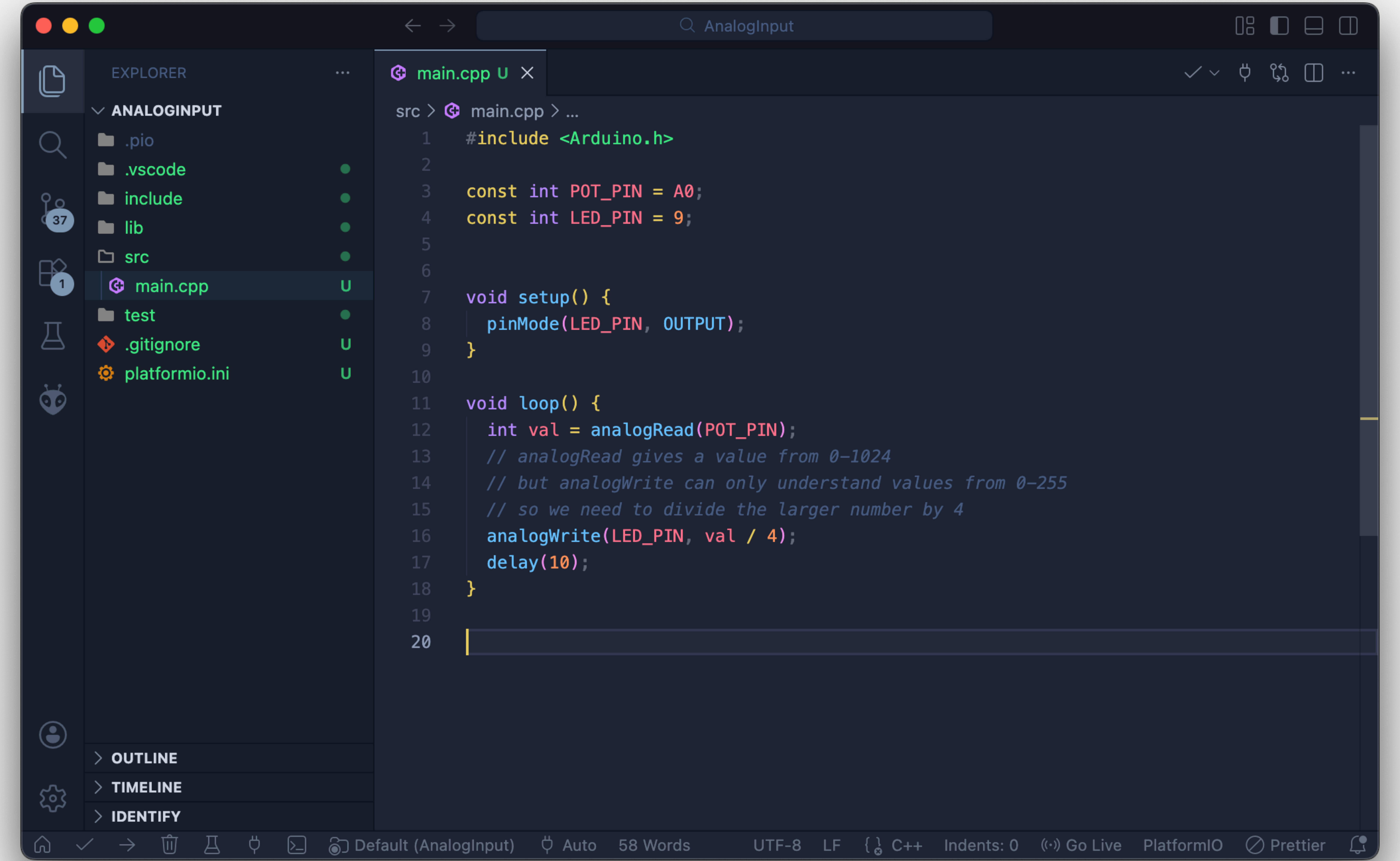
32u4 basic
adafruit!
RST 3V ARf GND A0 A1 A2 A3 A4 A5 SCK MOSI MISO RX TX GND
0 1
GND
leg2
Rotary Potentiometer (Small)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test
.gitignore U
platformio.ini U
OUTLINE
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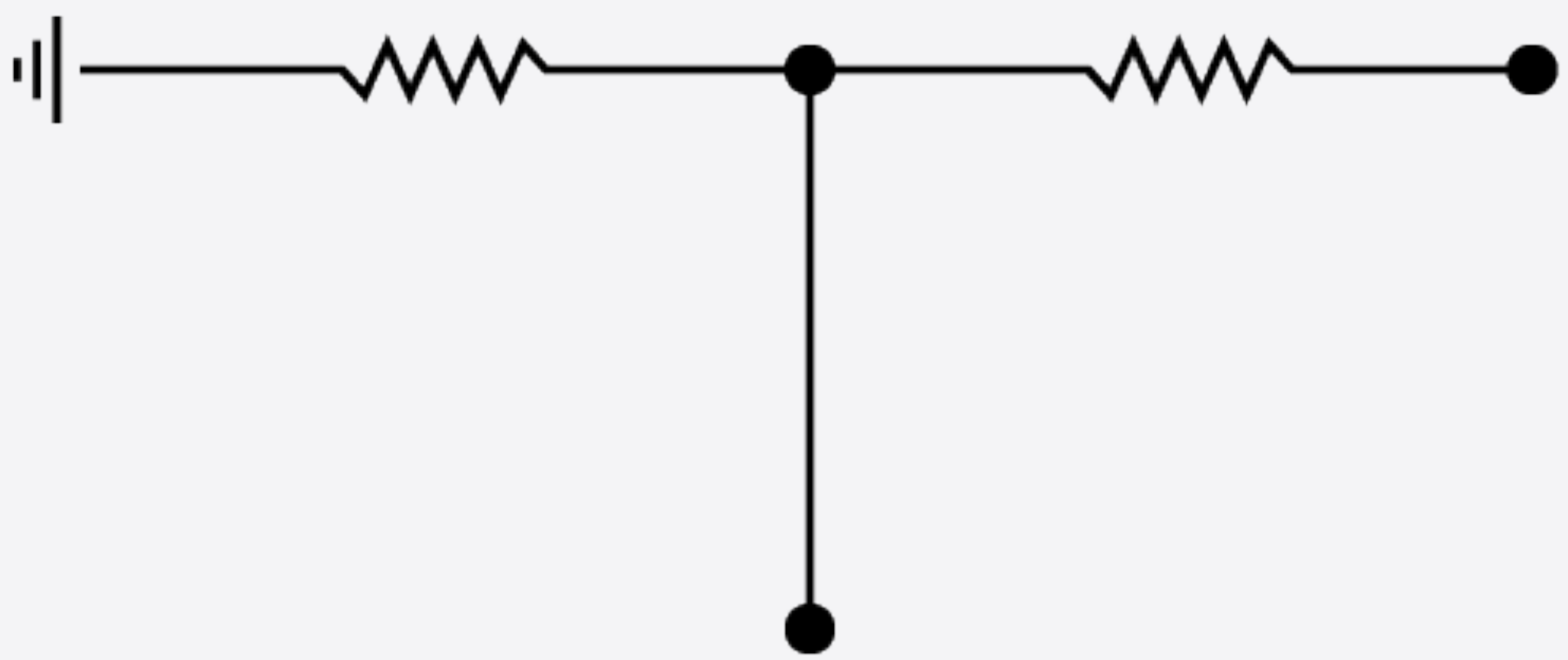
IDENTIFY
main.cpp U
src > main.cpp > ...
#include <Arduino.h>
const int POT_PIN = A0;
const int LED_PIN = 9;
void setup() {
pinMode(LED_PIN, OUTPUT);
}
void loop() {
int val = analogRead(POT_PIN);
// analogRead gives a value from 0-1024
// but analogWrite can only understand values from 0-255
// so we need to divide the larger number by 4
analogWrite(LED_PIN, val / 4);
delay(10);
}
Default (AnalogInput)
Auto
58 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier

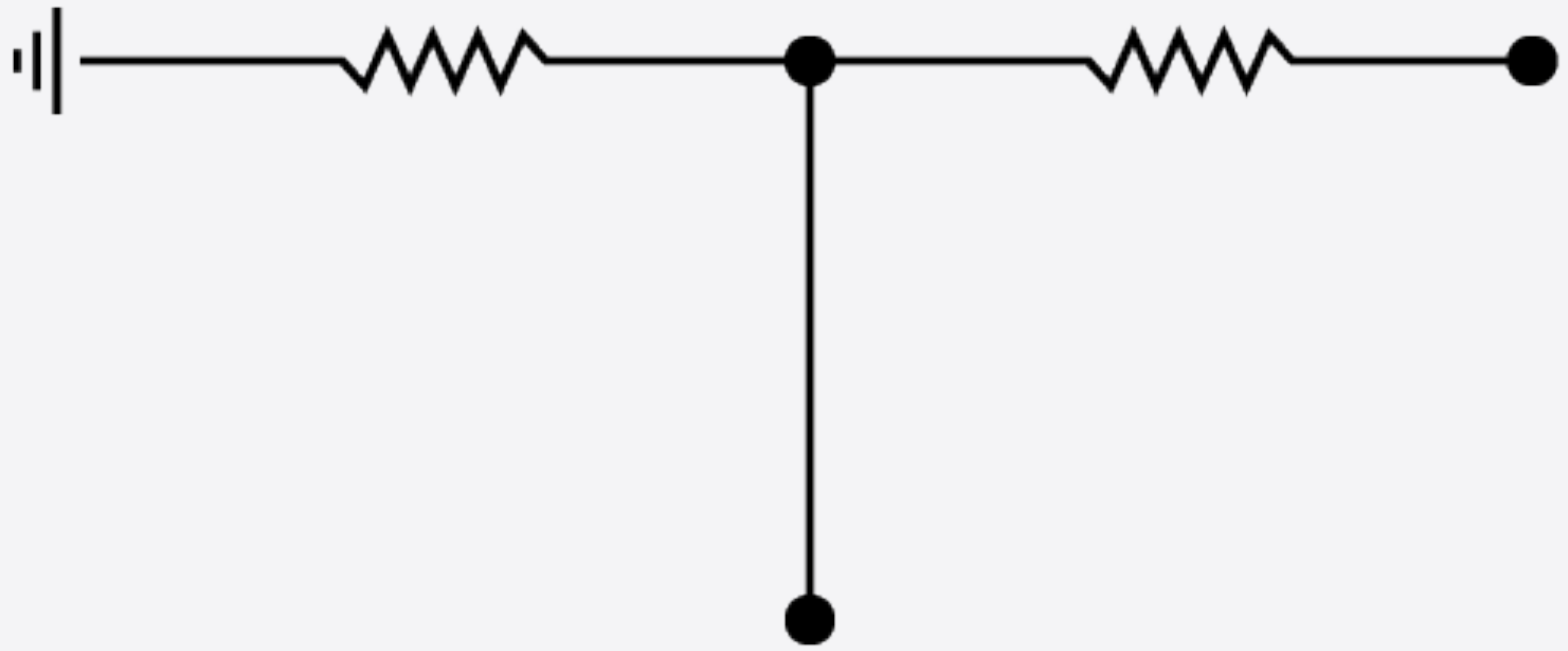
VOLTAGE DIVIDERS

Voltage Divider



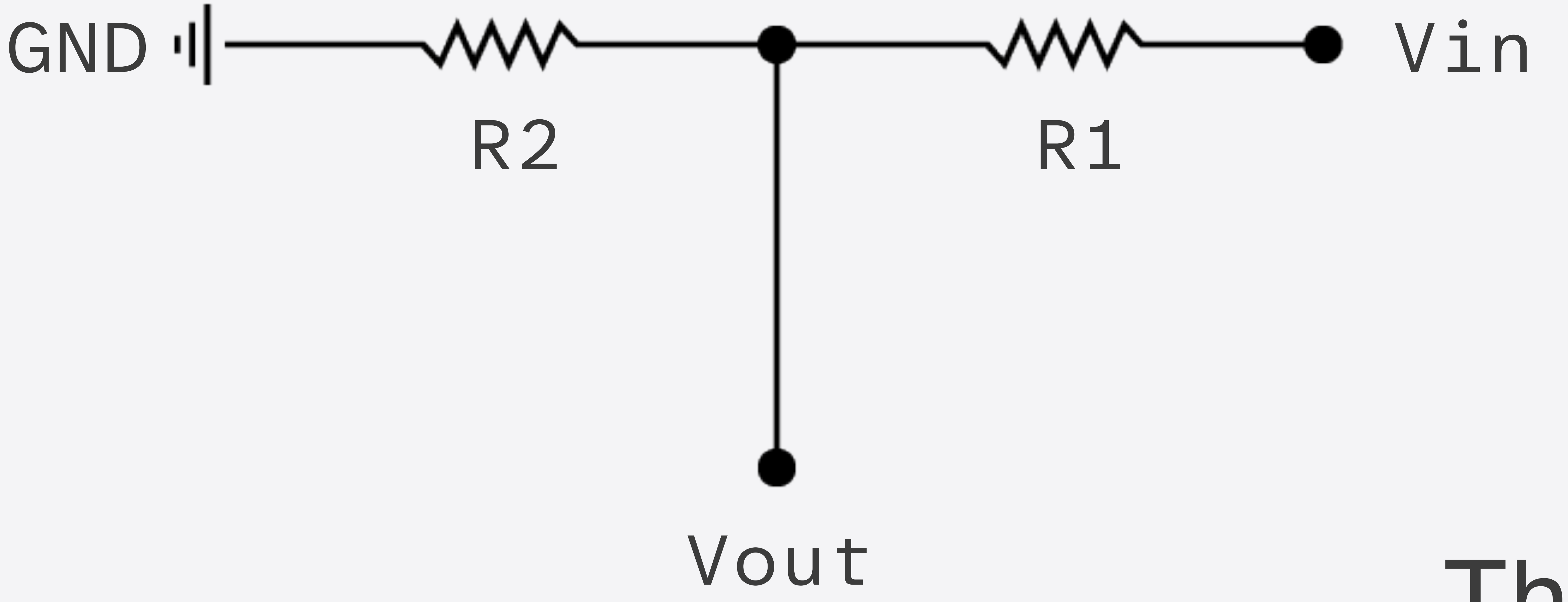
The Voltage Divider is a simple circuit that lets you convert a change in <u>resistance</u> into a change in <u>voltage</u>.

Voltage Divider



This is useful because there are a number of sensors that change resistance, but the Arduino can only sense changes in voltage.

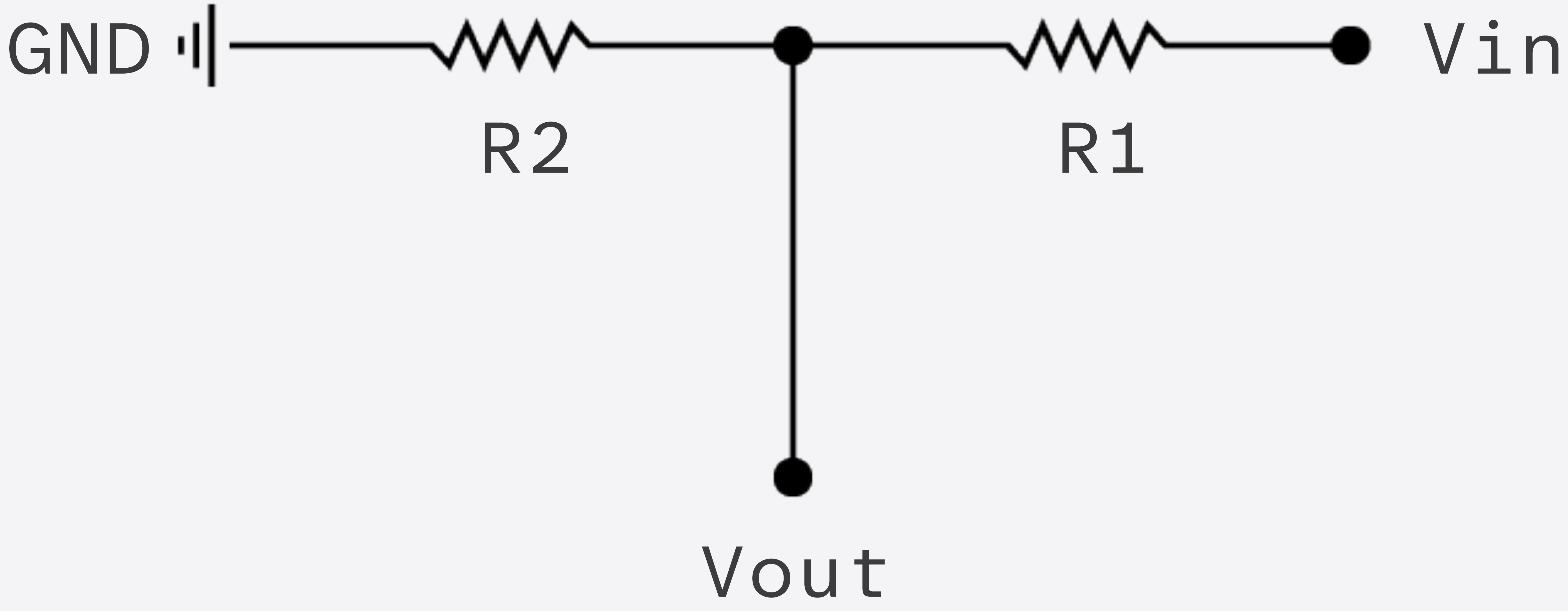
Voltage Divider



The resistor closest to ground is called R2.

The resistor closest to the input voltage (Vin) is called R1.

Voltage Divider



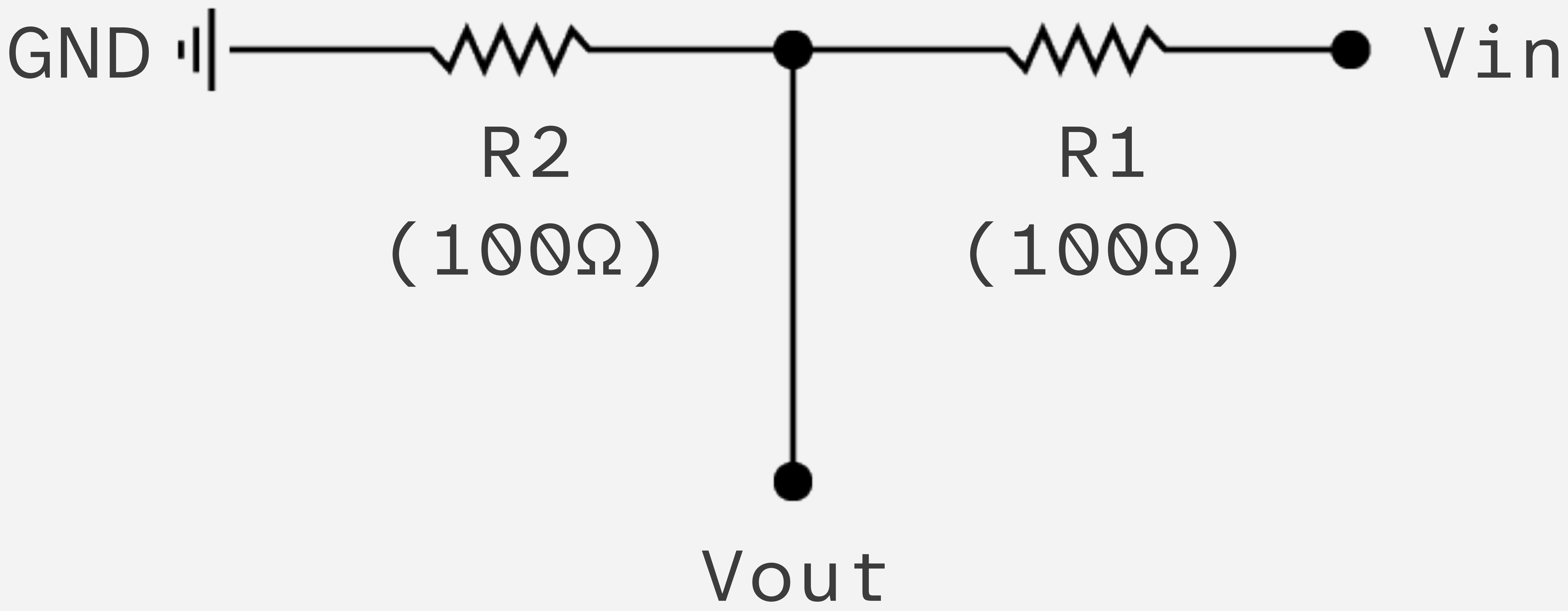
The result comes on `Vout`.

Voltage Divider

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

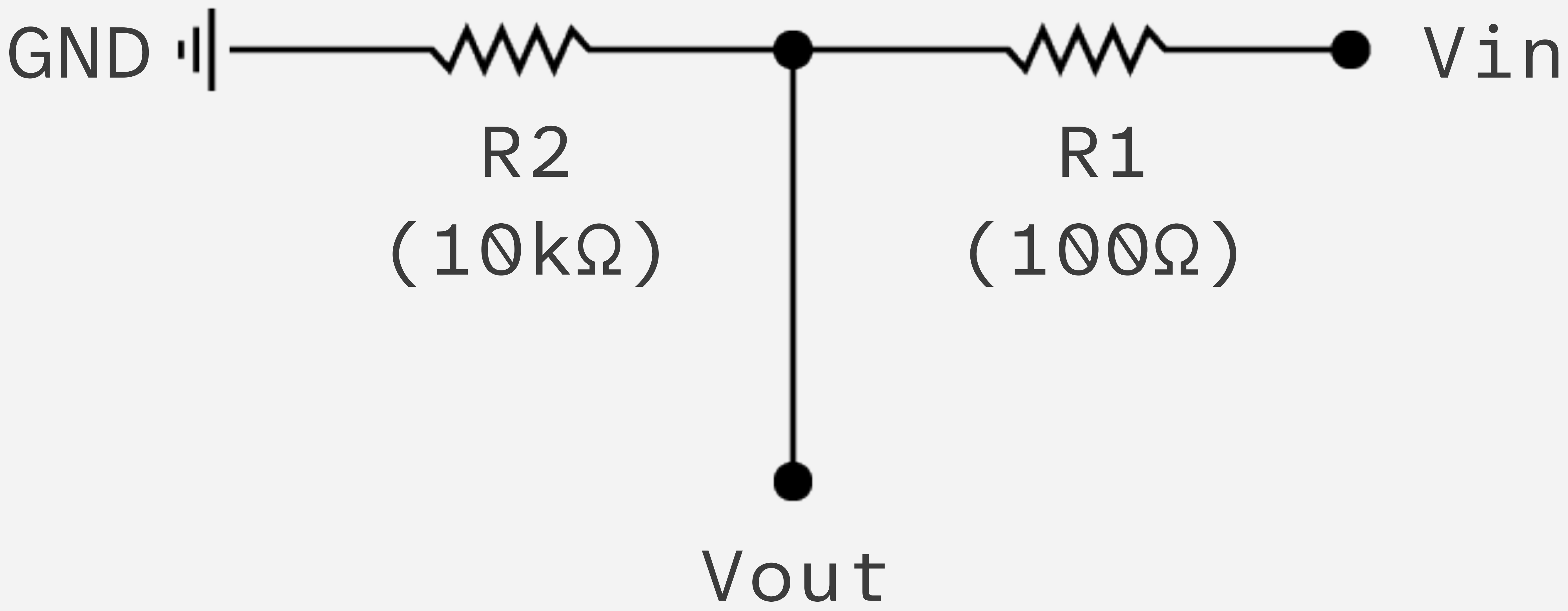
This equation lets you calculate the values.

Voltage Divider



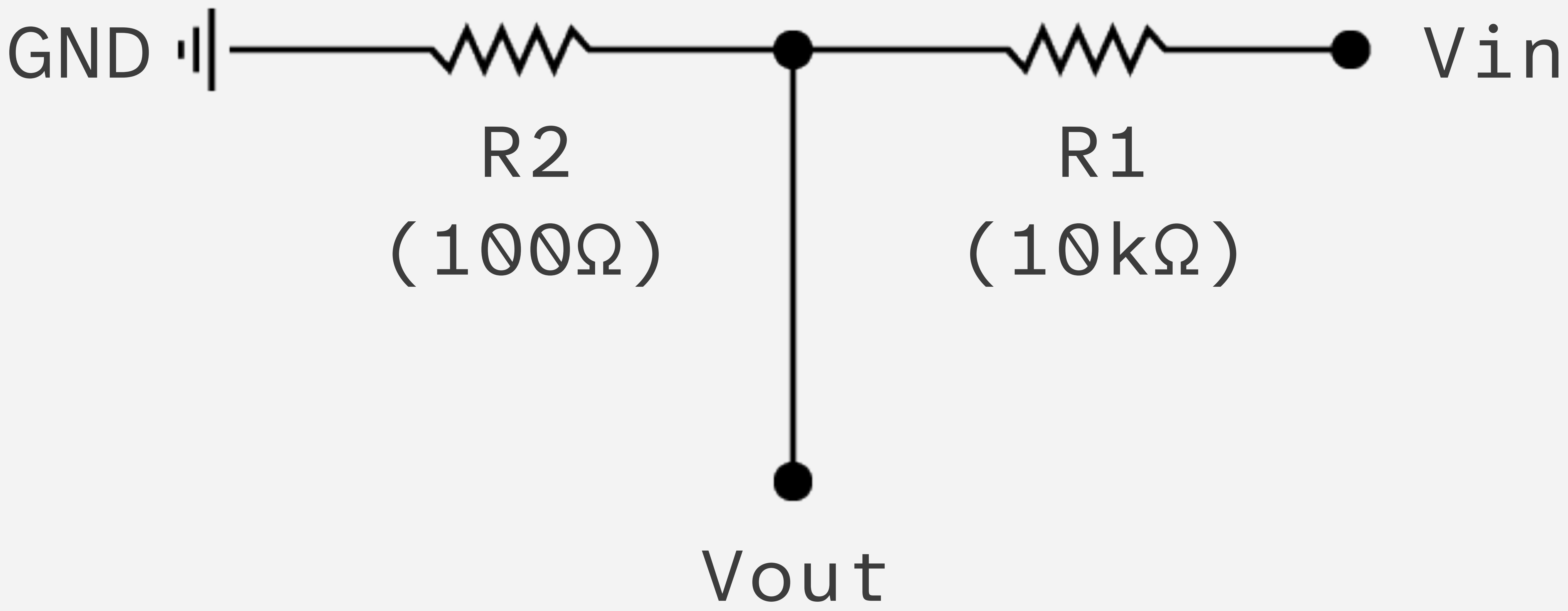
If R1 and R2 are equal
Then the output voltage is half
that of the input.

Voltage Divider



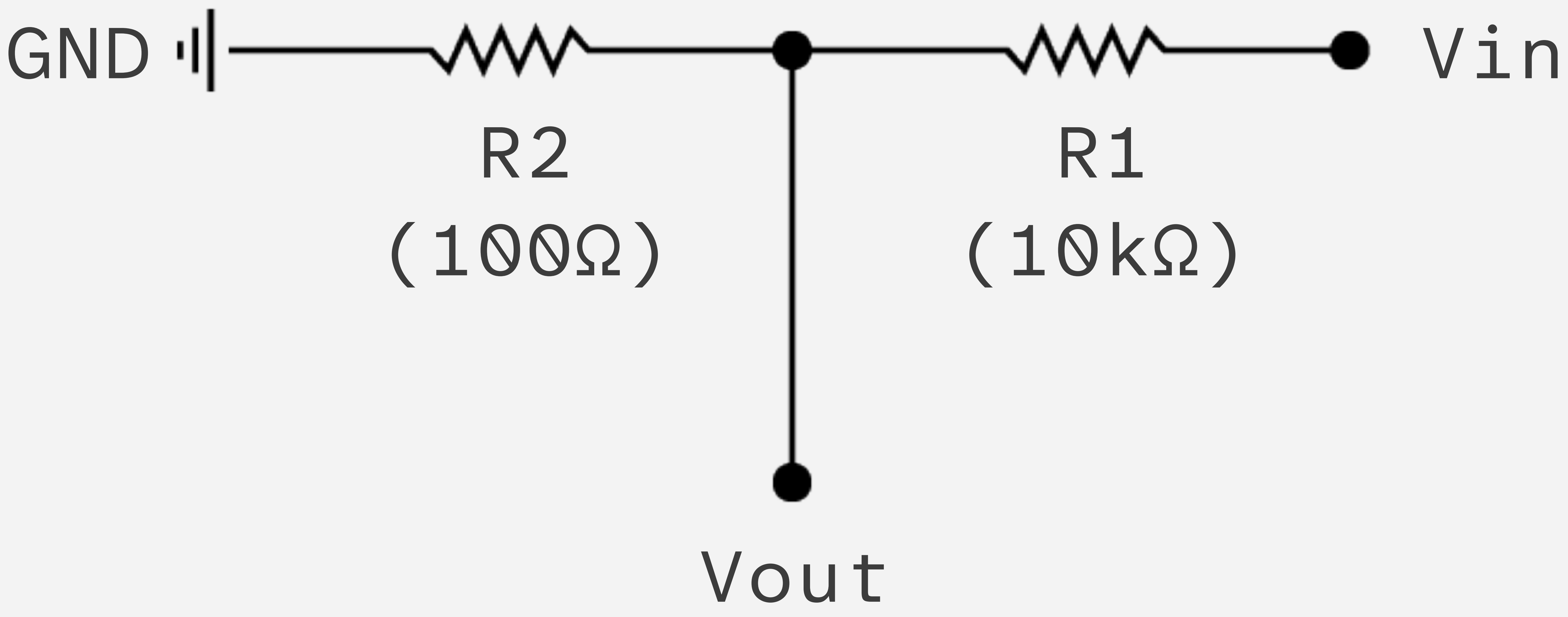
If R2 is much larger than R1
Then the output voltage will be
very close to the input.

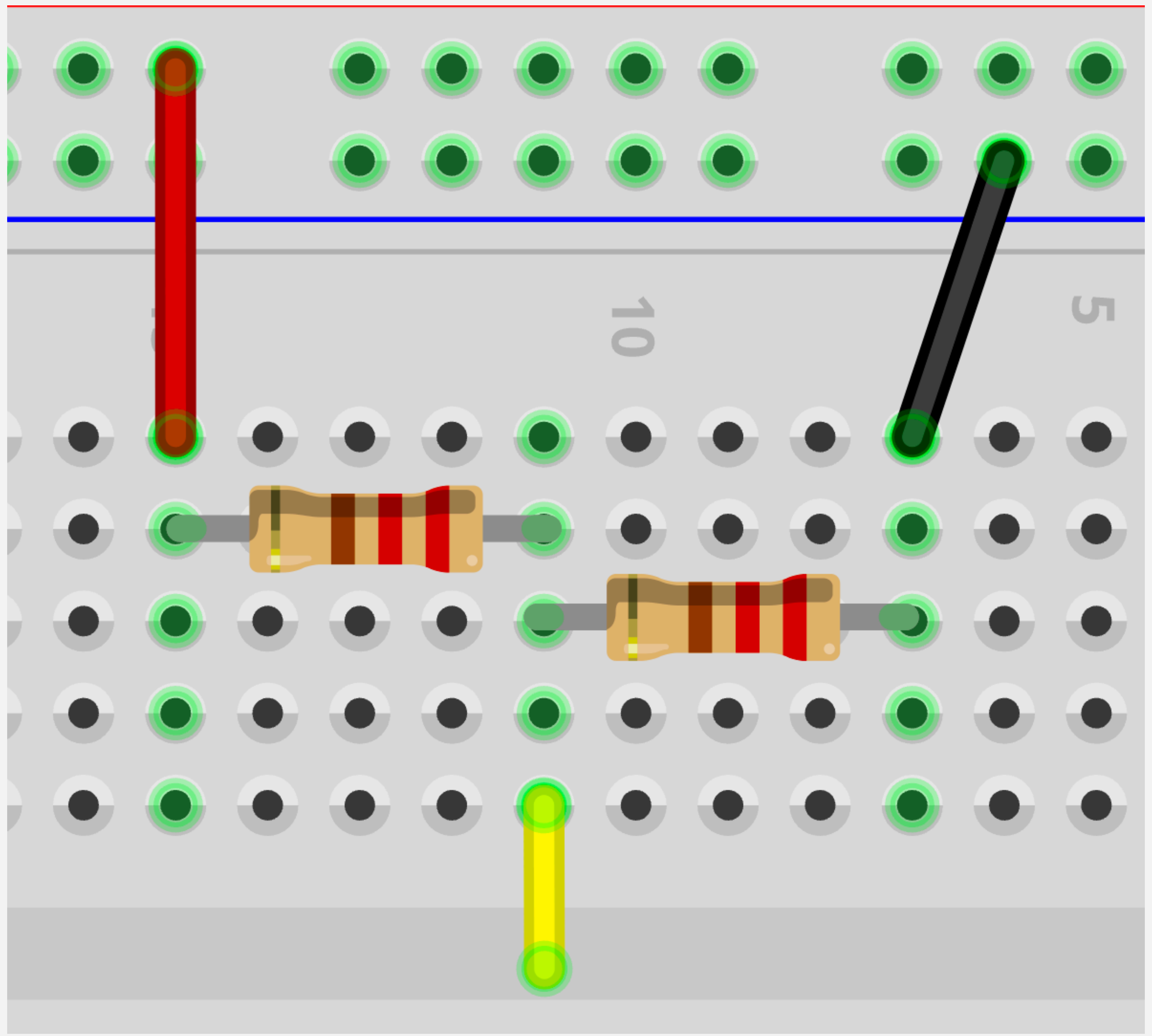
Voltage Divider



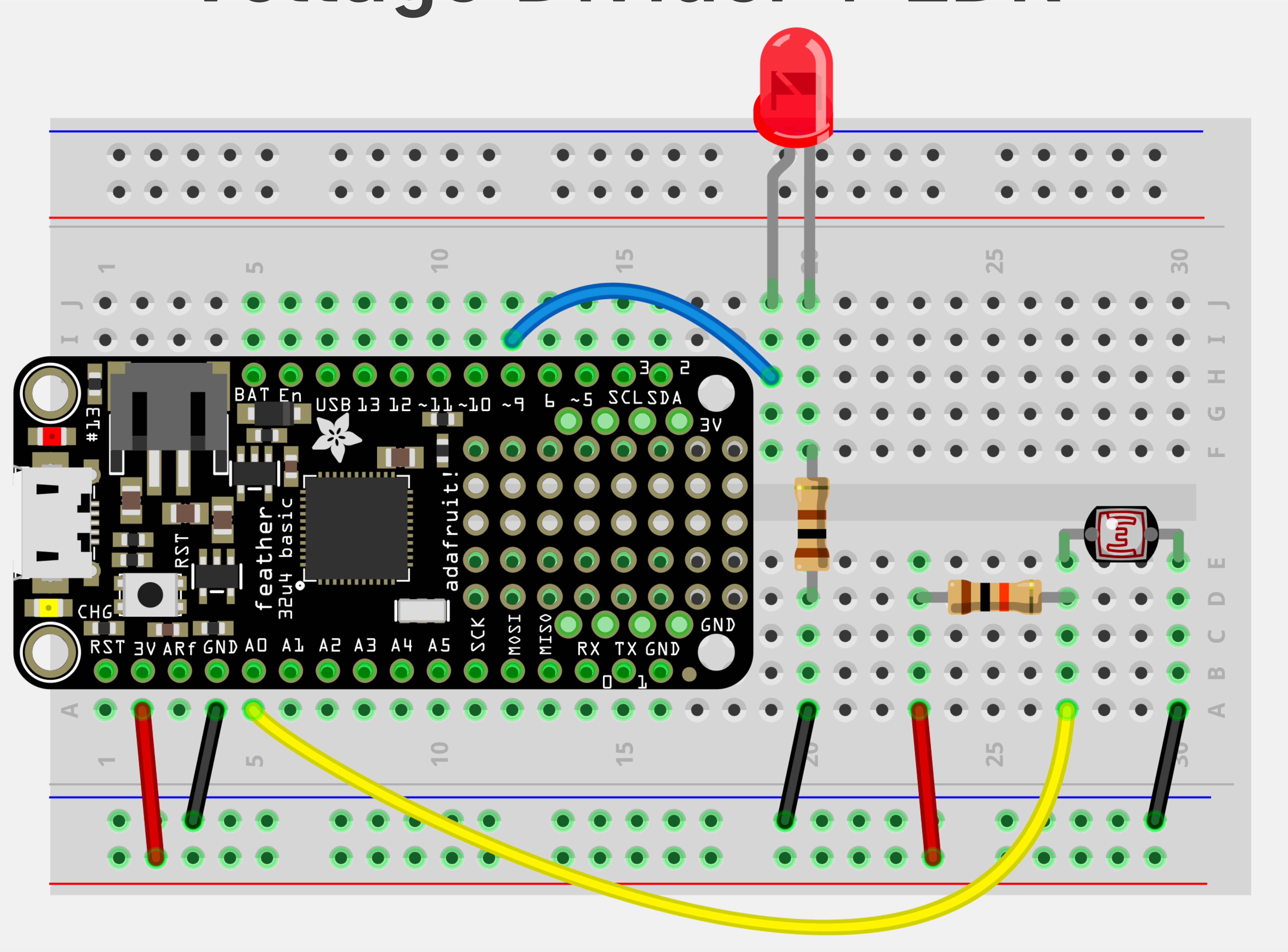
If R2 is much smaller than R1
Then output voltage will be tiny
compared to the input.

Voltage Divider

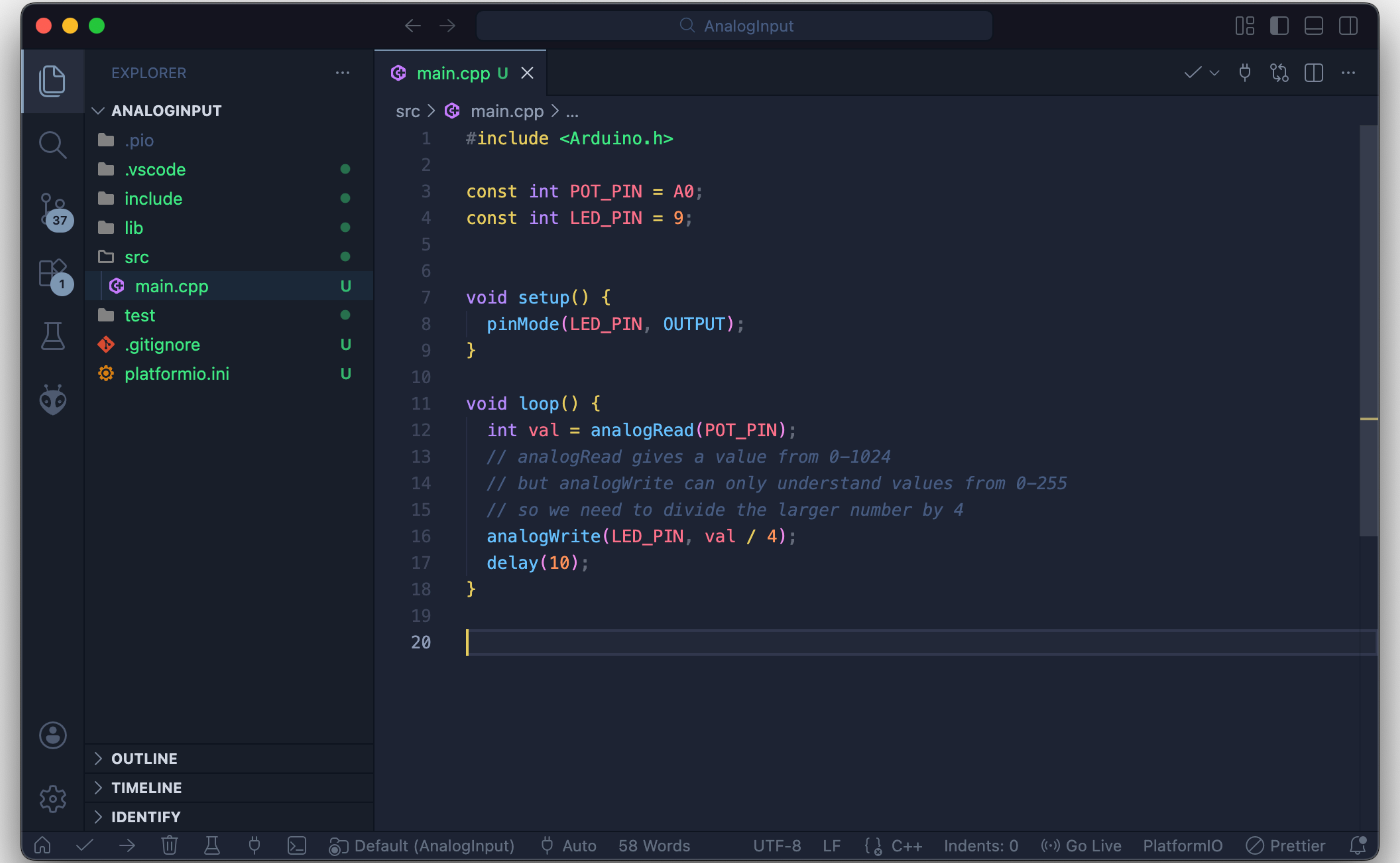




Voltage Divider + LDR*



* Light Dependent Resistor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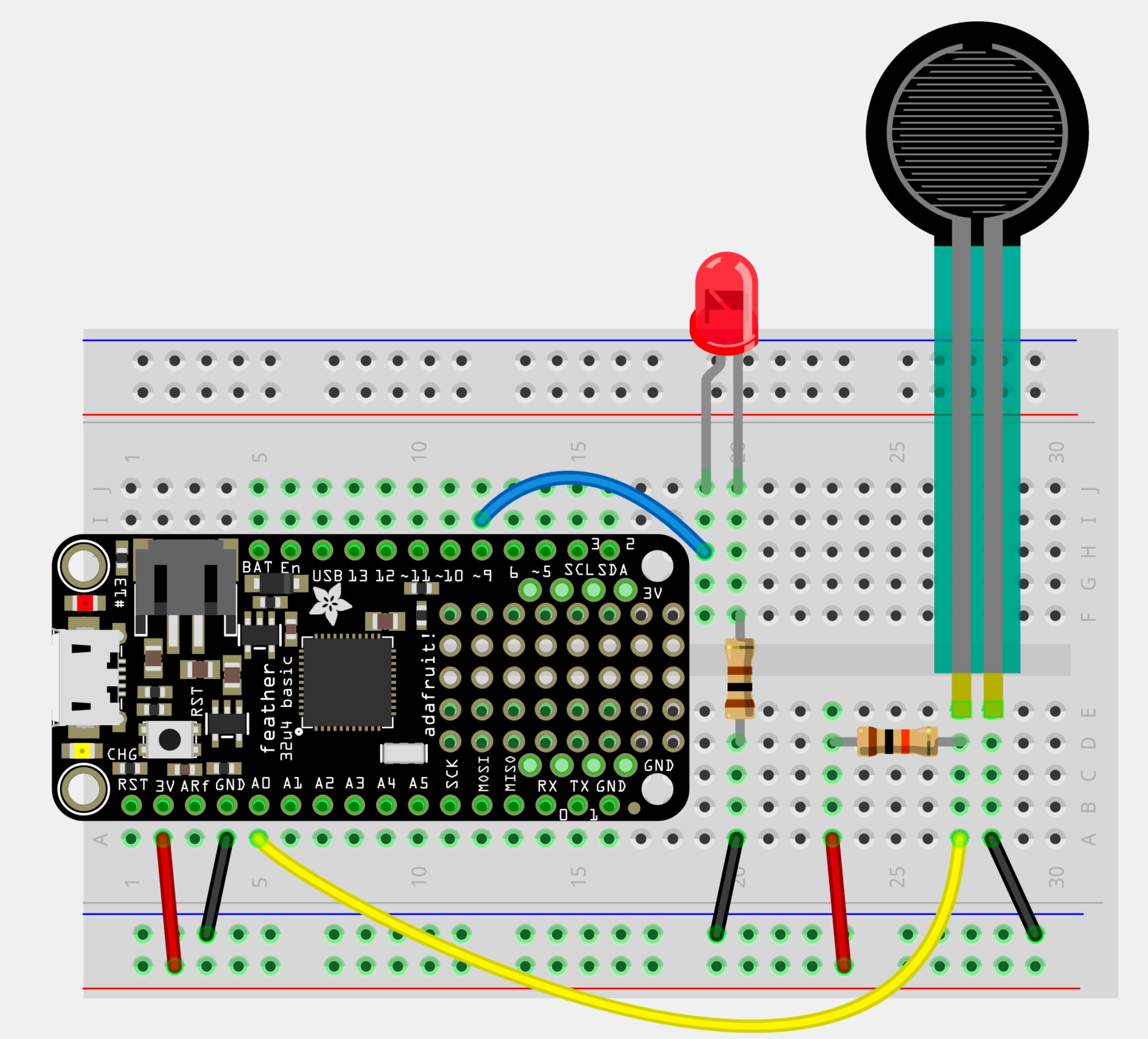
src > main.cpp > ...
#include <Arduino.h>
const int POT_PIN = A0;
const int LED_PIN = 9;
void setup() {
pinMode(LED_PIN, OUTPUT);
}
void loop() {
int val = analogRead(POT_PIN);
// analogRead gives a value from 0-1024
// but analogWrite can only understand values from 0-255
// so we need to divide the larger number by 4
analogWrite(LED_PIN, val / 4);
delay(10);
}
Default (AnalogInput)
Auto
58 Words
UTF-8
LF
C++
Indents: 0
Go Live
PlatformIO
Prettier

Voltage Divider + LDR*

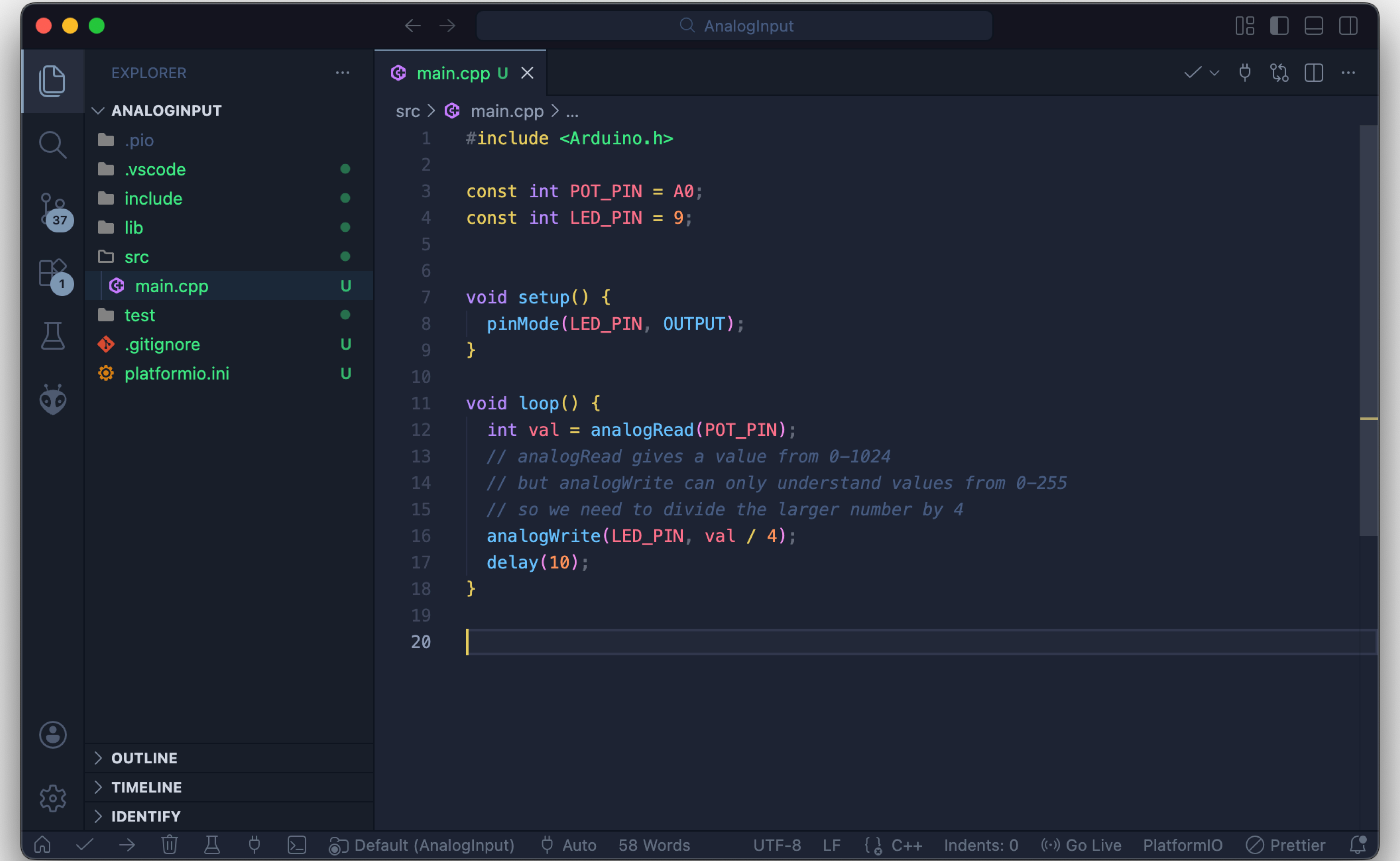
Light Level	R_2 (Sensor)	R_1 (Fixed)	$R_2/(R_1+R_2)$	V_{out}	Value from analogRead
Bright	300Ω	10kΩ	0.029	0.096 V	~30
Dim	7kΩ	10kΩ	0.411	1.35 V	~420
Dark	40kΩ	10kΩ	0.800	2.64 V	~819

* Light Dependent Resistor

Voltage Divider + FSR*



* Force Sensing Resistor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const int LED_PIN = 9;
void setup() {
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void loop() {
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}
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C++
Indents: 0
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